A Multiresolution 3D Morphable Face Model and Fitting Framework

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Abstract: 3D Morphable Face Models are a powerful tool in computer vision. They consists of a PCA model of face shape and colour information and allow to reconstruct a 3D face from a single 2D image. 3D Morphable Face Models are used for 3D head pose estimation, face analysis, face recognition, and, more recently, facial landmark detection and tracking. However, they are not as widely used as 2D methods - the process of building and using a 3D model is much more involved.

In this paper, we present the Surrey Face Model, a multi-resolution 3D Morphable Model that we make available to the public for non-commercial purposes. The model contains different mesh resolution levels and landmark point annotations as well as metadata for texture remapping. Accompanying the model is a lightweight open-source C++ library designed with simplicity and ease of integration as its foremost goals. In addition to basic functionality, it contains pose estimation and face frontalisation algorithms. With the tools presented in this paper, we aim to close two gaps. First, by offering different model resolution levels and fast fitting functionality, we enable the use of a 3D Morphable Model in time-critical applications like tracking. Second, the software library makes it easy for the community to adopt the 3D Morphable Face Model in their research, and it offers a public place for collaboration.

1 INTRODUCTION

3D face models and in particular 3D Morphable Models are a powerful tool for computer vision. They have applications in 2D face processing such as tracking, face analysis, recognition, pose estimation and pose normalisation. 3D Morphable Models (3DMM) were proposed by Blanz and Vetter in 1999 (Blanz and Vetter, 1999) and since then have been applied to a variety of these tasks. However they are not as widespread in use as their 2D counterparts (for example Active Appearance Models (Cootes et al., 2001)), yet they have certain distinct advantages over 2D methods. In a 3D model, the pose of a face is clearly separated from the shape. Its projection to 2D is modeled by a physical camera model, such as a perspective or affine camera model. Also, the use of a 3D face model allows to model the light explicitly since 3D surface normals, self-occlusion and depth information are available. The illumination model separates light from the face appearance and is not encoded in the texture parameters, as is for example the case in 2D AAMs. The most prominent approaches to modelling the environment or light sources in conjunction with 3DMMs are spherical harmonics (Aldrian and Smith, 2013; Zivanov et al., 2013) and the Phong illumination model (Romdhani and Vetter, 2005; Hu et al., 2012).

Furthermore, a 3D Morphable Model can be used in a generative way to create specific faces or to generate annotated training data for other algorithms that e.g. covers a large variety of pose angles, including more extreme poses like profile views. For example, Rätsch et al. (Rätsch et al., 2012) used 3DMM-generated data to improve the performance of a 2D pose regressor, while Feng et al. (Feng et al., 2015) augment the training data of their facial landmark detection with 3DMM data to make it more robust on larger pose angles. Very recently, 3DMMs have also been used directly with regression-based methods (Huber et al., 2015; Zhu et al., 2015) with aspirations to provide a unified solution to landmark detection and 3D model fitting.

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On the other hand, a 3D Morphable Face Model is comparatively hard to obtain and use. To train a model, a number of good quality 3D scans are needed. These scans then need to be brought into dense correspondence with a mesh registration algorithm. After building a model, a model-to-image fitting algorithm is required, and these fitting algorithms are often very complex, slow, and are easily trapped in local minima. The non-trivial training and fitting is in our opinion one of the main reasons for the limited adoption of 3D Morphable Face Models, and there is a lack of readily available 3D face model fitting frameworks.

In 2009, Vetter et al. published the Basel Face Model (BFM, (Paysan et al., 2009)) to spur research with Morphable Models. It surpassed existing models (Sarkar, 2005; Blanz and Vetter, 1999) by the accuracy of the scanner used and the quality of the registration algorithm, and their multi-segment face model, along with fitting results and various metadata, can be obtained after signing a licence agreement. While this led to some adoption by other research groups (e.g. (van Rootseler et al., 2012; Aldrian and Smith, 2013)), the adoption is still limited. Additionally, while the BFM provides the model, they only provide fitting results for limited databases and do not provide algorithms to apply the model to novel images. With the model and software framework published in this paper, we aim to go one step further and make our model available as well as a lightweight fitting framework promoting the use of the model.

This paper introduces the *Surrey Face Model* (SFM), a multi-resolution 3D Morphable Face Model. The model consists of three different resolution levels of shape and colour, a pose-invariant image texture representation and metadata such as landmark point information. The model is available freely for non-commercial purposes. Alongside the model, we present a library to interface with the model and perform basic pose and shape fitting. The main focus of the software is its ease of use and interoperability with the popular OpenCV library. The library is open source, available under *Apache License, Version 2.0*, and actively developed on GitHub, creating a public place for exchange and contributions. In addition to the full 3DMM being available via the University, we release a low-resolution shape-only model distributed within the software. In contrast to the Basel Face Model, we offer the models at lower resolutions as well, which are much more practical for many applications. We focus on ease of use of the framework for vision tasks such as face tracking, pose estimation and face frontalisation, and hope to further pave the way to make 3DMMs more widely adopted. Table 1 summarises how the different parts are available.

Table 1: Availability of the different components.

Component	**Availability & Licence**
Software library	GitHub, Apache License Version 2.0
Low-resolution shape model	GitHub, free for non-commercial purposes[†]
Full Surrey Face Model	After signing licence agreement, free for non-commercial purposes[†]

[†]*For commercial purposes, contact us via http://cvssp.org/facemodel.*

The contributions of this work are threefold. First, we present the Surrey Face Model, a multi-resolution 3D Morphable Face Model that we make available to the public for non-commercial purposes. Second, we present a lightweight open-source Morphable Model software framework written in modern C++ that is designed with simplicity and ease of integration as its primary goals. Lastly, we make a low resolution shape model available together with the software to allow an immediate start with the framework. With this whole package, our aim is to make 3D Morphable Models easily available and to encourage research with 3D face models.

In the rest of this paper, we first describe the acquisition and the building process as well as the metadata of the Surrey Face Model in detail (Section 2). We introduce the resolution levels and show visualisations of the model and its PCA components. In Section 3, we then present the accompanying software framework and demonstrate its ease of use and flexibility by means of a 3D pose estimation and face frontalisation example. Section 4 concludes the paper.

2 THE SURREY FACE MODEL

The Surrey Face Model consists of a 3D Morphable Model, that is, a PCA shape model and a PCA colour model, each in different resolution levels, and accompanying metadata, like a 2D texture representation and landmark annotations. The following sections will describe each part in detail.

2.1 3D Morphable Models

A 3D Morphable Model is based on three dimensional meshes of faces that have been registered to a reference mesh, i.e. are in dense correspondence. A face is represented by a vector $S \in \mathbb{R}^{3N}$, containing the x, y and z components of the shape, and a vector $T \in \mathbb{R}^{3N}$, containing the per-vertex RGB colour information. N is the number of mesh vertices. The 3DMM consists of two PCA models, one for the shape and one for the

colour information. Each PCA model

$$M := (\bar{\mathbf{v}}, \sigma, \mathbf{V}) \tag{1}$$

consists of the components $\bar{\mathbf{v}} \in \mathbb{R}^{3N}$, which is the mean of the example meshes, a set of principal components $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}] \in \mathbb{R}^{3N \times n-1}$, and the standard deviations $\sigma \in \mathbb{R}^{n-1}$. n is the number of scans used to build the model. Novel faces can be generated by calculating

$$S = \bar{\mathbf{v}} + \sum_{i}^{M} \alpha_i \sigma_i \mathbf{v}_i \tag{2}$$

for the shape, where $M \leq n-1$ is the number of principal components and $\alpha \in \mathbb{R}^M$ are the 3D face instance coordinates in the shape PCA space. The same is calculated for the colour (or so-called albedo) model.

2.2 3D Scan Data

The Surrey Face Model is built using a number of high-resolution 3D scans that were acquired at our lab. The scans were captured using a 3dMDface[2] camera system that consists of two structured light projectors, 4 infrared cameras that capture the light pattern and are used to reconstruct the 3D shape, and two RGB cameras recording a high-resolution face texture. Half the cameras record the face from the left side, the other half from the right side, resulting in a 180° view of the face. The images are acquired under uniform illumination to ensure that the model texture is representative of face skin albedo only. The 3dMDface software reconstructs a textured 3D mesh from this information.

These scans can then be brought into dense correspondence using a 3D to 3D registration algorithm, in our case the Iterative Multi-resolution Dense 3D Registration (IMDR) method (Tena et al., 2006). Figure 1 shows an example scan with the captured mesh on the left, the RGB texture in the middle, and the scan after registration to the 3D model. The registration process is described in more detail in the next section together with how we built the multi-resolution model.

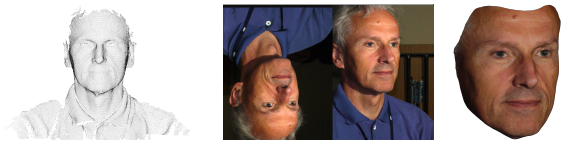


Figure 1: *(left)*: Raw mesh output from the 3dMDface software. *(middle)*: Texture image from two angles captured by the 3dMDface cameras. *(right)*: The scan and texture densely registered to the 3D model.

[2] http://www.3dmd.com/

Our recorded subjects represent a diverse range of skin tones and face shapes to well represent the multicultural make up of many modern societies. Figure 2 shows the perceived racial distribution of the 169 scans used to build the model. Non-Caucasian people are well-represented and significant numbers of subjects from other races are included allowing the model to generalise well to people from various backgrounds. This is in stark contrast to the BFM, which only contains a very insignificant number of non-Caucasian people.

The age of the recorded people was categorised into 5 groups. 9 are teens (age 0-19), 106 young adults (20-29), 33 adults (30-44), 13 mature (45-59) and 8 seniors (60+). Similar to the BFM, most people are in the young-adult range, but the Surrey Face Model contains more people in the 30+ groups.

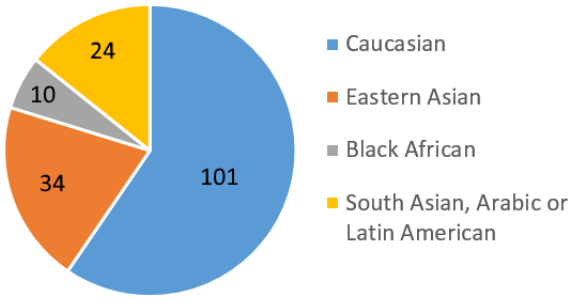


Figure 2: Racial distribution of the 169 scans used to train the Surrey Face Model. Our model is generated from a significant number of non-Caucasian people.

2.3 Multi-Resolution Model Generation

The Surrey Face Model comes in three different resolution levels. After obtaining the full set of scans, they are registered using the Iterative Multi-resolution Dense 3D Registration (IMDR) algorithm. It uses a deformable reference 3D face model and performs a combination of global mapping, local matching and energy-minimisation to establish dense correspondence among all the scans at different resolution levels. The generic reference face we used has 845 vertices and 1610 triangles. The following is a high-level overview of the process:

1. The target scan is denoised using Gaussian and median filtering if spikes and noise are present.
2. Perform a global mapping from the generic model to the target scan using facial landmarks, smoothly deforming the generic model.
3. Do a local matching on the current resolution level based on the distances between reference and target vertices. If a particular vertex cannot

be matched, its mirrored counterpart is used (and if that fails as well, the algorithm interpolates using the neighbouring matches).

4. The final set of matches guides an energy minimisation process that conforms the model to the target scan. Steps 3 and 4 are iterated.
5. The generic face model is subdivided using the 4-8 mesh subdivision algorithm (Velho and Zorin, 2001).
6. Steps 3 to 5 are repeated until the desired highest mesh resolution is achieved.

Table 2 shows the three constructed resolution levels with their number of vertices and triangles. The smallest model consists of 3448 vertices and the full model consists of 29587 vertices. Figure 3 depicts the three mesh resolutions with a close-up on the model's mesh. Note that the higher resolution meshes are built upon the lower resolutions, and therefore each vertex from a lower resolution mesh is also present in all higher resolution meshes, and has the same vertex index.

Table 2: The different model resolution levels.

Model name	No. vertices	No. triangles
sfm_29587	29587	59763
sfm_16759	16759	33211
sfm_3448	3448	6736

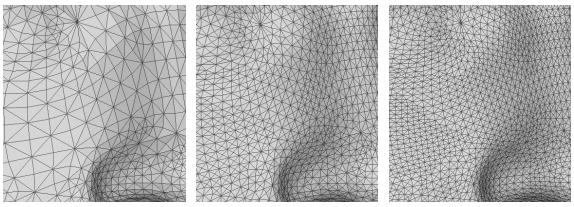


Figure 3: Close-up of the different mesh resolutions of the Surrey Face Model. *(left)*: The low-resolution model with 3448 vertices. *(middle)*: Medium-resolution model (16759 vertices) *(right)*: The full resolution model (29587 vertices).

2.4 Shape and Colour Model

The shape and colour PCA models are built using aforementioned 169 registered scans. Of the resulting PCA basis matrix, we keep 63 shape eigenvectors and 132 colour eigenvectors so that 99% of the original variation of the data is preserved after reconstruction. To analyse the variations in the face model, we can visualise the directions of largest variance in the PCA space by taking the formula in Equation 2 and setting a specific α_i to a fixed value while setting all others to zero. The resulting face mesh S can then be rendered. Figure 4 shows the mean of the model

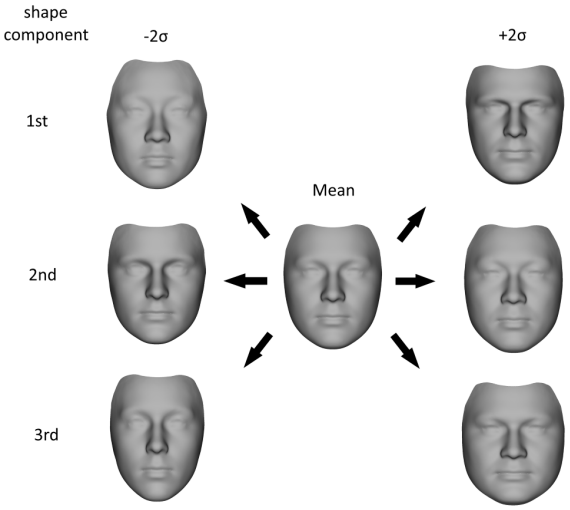


Figure 4: The mean face and shape variation of the high-resolution Surrey Face Model. The figure shows the first three PCA shape coefficients at -2 and +2 standard deviations.

and the first three shape components set to ± 2 standard deviations. The first components mainly account for global structure of the face, like global face shape (more round or square, slim or chubby) and size of the face. Later components model the finer structures of the face.

Figure 5 depicts the colour PCA model with the colour coefficients set to ± 2 standard deviations. Varying the first component of the colour model results mainly in a change of global skin colour from black to white, while the second component models more diverse changes relating to the gender. The third component encodes a mixture of skin colour and possibly gender.

Along with the model, we publish annotations of the most commonly used facial landmark points for all resolution levels. When using the model, it is useful to have a standardised and known set of the location (i.e. vertex index) of certain points like the eye corners or the tip of the nose on the mesh. We provide such metadata with the model. Whenever possible, the points are defined on the generic reference face or in the lowest model resolution and valid on all model levels. Some points are only defined on the higher mesh resolutions because the mesh resolution at a lower level is too coarse and does not have a vertex at the landmark location. Figure 6 shows a set of manually selected landmark points on the mesh that correspond to a subset of the popular ibug facial point annotations[3].

[3] http://ibug.doc.ic.ac.uk/resources/facial-point-annotations/

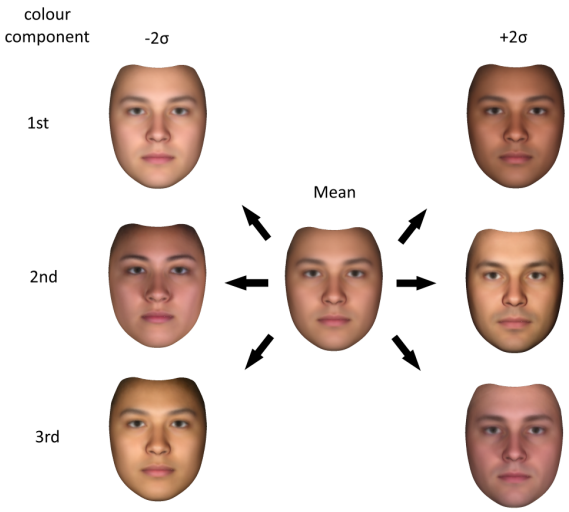


Figure 5: The mean face and colour variation of the high-resolution Surrey Face Model. The figure shows the first three PCA colour coefficients at -2 and +2 standard deviations.

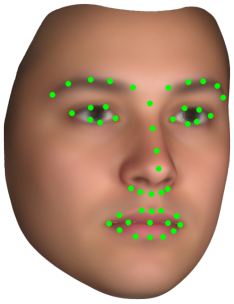


Figure 6: The facial landmark points that are annotated on the mesh and available as metadata together with the model.

2.5 Texture Representation

The PCA colour model is a useful representation for the appearance of a face, but in some cases it is desirable to use the pixel colour information (*texture*) from the image or a combination of the two. The texture from the input image remapped onto the mesh preserves all details of a face's appearance, while some high-frequency information can be lost if a face is only represented using the PCA colour model. Another reason to use the texture is to avoid a colour and light model fitting, for example in consideration of run-time. Therefore, we would like a 2D representation of the whole face mesh that we can use to store the remapped texture. We create such a generic representation with the isomap algorithm (Tenenbaum et al., 2000): it finds a projection from the 3D vertices to a 2D plane that preserves the geodesic distance between the mesh vertices. Our mapping is computed with the algorithm from Tena (Rodríguez, 2007).

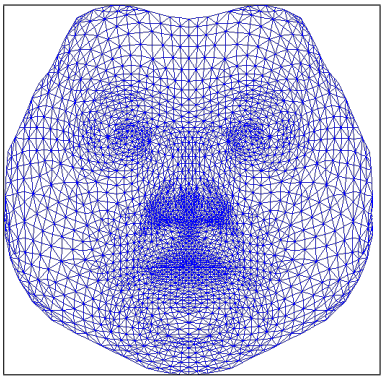


Figure 7: Texture representation in the form of an isomap. The 3D mesh vertices are projected to 2D with an algorithm that preserves the geodesic distance between vertices, resulting in a pose-independent, detail-preserving textural representation of a face. Shown is the isomap of the *sfm_3448*.

In contrast to other representations, like for example cube mapping, this isomap has the advantage that it can be stored as a single 2D image, and it has face-like appearance, i.e. it can be easily used with existing face recognition and face analysis techniques. The isomap coordinates are only generated once, that is the isomaps of different people are in dense correspondence with each other, meaning each location in the map corresponds to the same physical point in the face of every subject (for example, a hypothetical point $x = [100, 120]$ is always the center of the right eye). This makes the isomap especially suitable for processing with further algorithms.

Figure 7 shows the isomap of the low-resolution model as a wireframe. The isomap captures the whole face, while for example a rendering of the mesh would always only show parts of the face.

We provide texture coordinates generated with the isomap algorithm for each model resolution level.

3 SOFTWARE FRAMEWORK

The Surrey Face Model is accompanied by an open source software framework that is available on GitHub under *Apache License, Version 2.0*. It is a lightweight and cross-platform header-only library built using modern C++, which makes it very flexible, easy to use and simple to include into existing software. The library requires no other dependency than applications developed using it linking to OpenCV-core. The software framework includes a low-resolution shape-only model (*sfm_shape_3448*) to facilitate immediate use.

This section will first give a brief introduction about the core functionality of the framework and then present the pose and landmark fitting algorithms included in the software and a basic use case example. The library, low-resolution shape model, example applications and a complete documentation are available at `https://github.com/patrikhuber/eos`.[4]

3.1 Core Functionality

A 3D Morphable Model is represented using a thin layer on top of OpenCV. A `PcaModel` class contains a mean, the PCA basis vectors and eigenvalues, and a `MorphableModel` class contains a shape and colour PCA model and texture coordinates, along with the functionality to retrieve a mesh that can be rendered. Figure 8 shows an UML-like overview of this basic functionality.

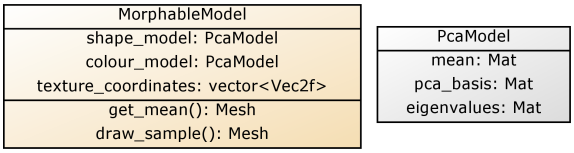


Figure 8: UML-like diagram of the `MorphableModel` and `PcaModel` classes, which are simple wrappers for the underlying PCA and model metadata (diagram slightly simplified).

A model can be loaded as simple as:

```
MorphableModel model =
 ↪  load_model("sfm_shape_3448.bin");
```

The semantic landmark information (see end of Section 2.4) is stored separately as it differs depending on the use-case and it can be accessed through a `LandmarkMapper` class. In the following, we will extend this code snippet and build a 3D face frontalisation example with the software framework in only a few lines of code.[5]

In addition to loading the Surrey Face Models, we provide a script to convert and load the Basel Face Model.

3.2 Landmark Fitting

The library includes methods to fit the pose and shape of a model and perform face frontalisation. This section describes the individual components and how they fit together.

3.2.1 Pose Estimation

The first component presented here is pose (camera) fitting. Given a set of 2D landmark locations and their known correspondences in the 3D Morphable Model, the goal is to estimate the pose of the face (or the position of the camera, which in this case is the identical problem). We assume an affine camera model and implement the *Gold Standard Algorithm* of Hartley & Zisserman (Hartley and Zisserman, 2004), which finds a least squares approximation of a camera matrix given a number of 2D - 3D point pairs.

First, the detected or labeled 2D landmark points in the image $x_i \in \mathbb{R}^3$ and the corresponding 3D model points $X_i \in \mathbb{R}^4$ (both represented in homogeneous coordinates) are normalised by similarity transforms that translate the centroid of the image and model points to the origin and scale them so that the Root-Mean-Square distance from their origin is $\sqrt{2}$ for the landmark and $\sqrt{3}$ for the model points respectively: $\tilde{x}_i = \mathbf{T}x_i$ with $\mathbf{T} \in \mathbb{R}^{3\times3}$, and $\tilde{X}_i = \mathbf{U}X_i$ with $\mathbf{U} \in \mathbb{R}^{4\times4}$. Using ≥ 4 landmark points, we then compute a normalised camera matrix $\tilde{\mathbf{C}} \in \mathbb{R}^{3\times4}$ using the *Gold Standard Algorithm* (Hartley and Zisserman, 2004) and obtain the final camera matrix after denormalising: $\mathbf{C} = \mathbf{T}^{-1}\tilde{\mathbf{C}}\mathbf{U}$.

Computing the camera matrix $\mathbf{C}$ involves solving a linear system of equations - the algorithm calculates the least squares solution, so any number of corresponding points can be given. This process is also very fast, taking only a few milliseconds to compute.

The function in our library can be run by simply loading and defining point correspondences and then calling the algorithm:

```
vector<cv::Vec2f> image_points = ...;
vector<cv::Vec3f> model_points = ...;
Mat affine_camera =
 ↪  estimate_affine_camera(image_points,
 ↪  model_points);
```

which returns the estimated 3×4 affine camera matrix that can subsequently be used in the next steps.

3.2.2 Shape Fitting

The second component in our face frontalisation example consists of reconstructing the 3D shape using the estimated camera matrix. We implement a simple shape-to-landmarks fitting similar to the algorithm from Aldrian & Smith (Aldrian and Smith, 2013). We find the most likely vector of PCA shape coefficients

[4] The functionality described in this paper is based on *v0.5.0*. Naturally, software evolves, and certain parts might be slightly different in future versions. We encourage users to look at the example app of the most current version.

[5] For the sake of brevity, we omit namespaces in the code examples. Where not otherwise indicated, types and functions are either in the `std` or our library's namespace.

α by minimising the following cost function:

$$\mathbb{E} = \sum_{i=1}^{3N} \frac{(y_{m2D,i} - y_i)^2}{2\sigma_{2D}^2} + \|\alpha\|_2^2, \qquad (3)$$

where N is the number of landmarks, y are detected or labelled 2D landmarks in homogeneous coordinates, σ_{2D}^2 is an optional variance for these landmark points, and y_{m2D} is the projection of the 3D Morphable Model shape to 2D using the estimated camera matrix. More specifically, $y_{m2D,i} = \mathbf{P}_i \cdot (\hat{\mathbf{V}}_h \alpha + \bar{\mathbf{v}})$, where $\mathbf{P}_i$ is the i-th row of $\mathbf{P}$ and $\mathbf{P}$ is a matrix that has copies of the camera matrix $\mathbf{C}$ on its diagonal, and $\hat{\mathbf{V}}_h$ is a modified PCA basis matrix that consists of a sub-selection of the rows that correspond to the landmark points that the shape is fitted to. Additionally, a row of zeros is inserted after every third row to accommodate for homogeneous coordinates, and the basis vectors are multiplied with the square root of their respective eigenvalue. The cost function in (3) can be brought into a standard linear least squares formulation. For details of the algorithm, we refer the reader to (Aldrian and Smith, 2013).

This functionality is directly mapped into code. The shape coefficients can be estimated as:

```
vector<float> shape_coefficients =
 ↪  fit_shape_to_landmarks_linear(model,
 ↪  affine_camera, image_points,
 ↪  vertex_indices, lambda);
```

The first three parameters are given from the previous step. The `vertex_indices` are obtained with the supplied landmark annotation metadata and the mapping facilities of the library - we refer to the online material for the full documentation. `lambda` is an optional regularisation parameter to constrain the optimisation to plausible shapes.

The pose estimation and shape fitting steps can be iterated if desired to refine the estimates. The pose estimation can make use of the shape estimate (instead of using the mean face) to refine the face pose. The shape estimate can in turn use the refined camera matrix to improve the shape fitting. The shape estimation is as fast as the pose estimation: each of them only involves solving a small linear system of equations and runs in the order of milliseconds.

3.2.3 Texture Representation

After obtaining the pose and shape coefficients, there is a dense correspondence between mesh vertices and the face in the input image. We can then remap the texture onto the model, and store it, and re-render it in arbitrary poses, e.g. frontalise it. The texture can be extracted and stored in the isomap introduced in Section 2.5 with another simple call to the library:

```
Mesh model_instance =
 ↪  model.draw_sample(shape_coefficients);
Mat isomap = extract_texture(model_instance,
 ↪  affine_camera, input_image);
```

Note that, in addition to being used to remap the texture or visualise the model, `draw_sample` allows drawing arbitrary samples from the model, for example to generate artificial training data from the 3D face model.

Figure 9 shows an example fitting with the input image, the resulting shape and camera model fitting, and the extracted face texture as an isomap. In the figure, regions of self-occlusion are depicted as white spots; however, in the isomap, they are identified by the alpha channel.



Figure 9: An example result of the landmark fitting. *(left)*: Input image from LFPW. *(middle)*: The extracted face texture as isomap. Regions of self-occlusion are depicted in white. *(right)*: Resulting shape and camera model fitting.

This section presented how the proposed lightweight library makes it possible to build a real-time 3D pose estimation and face frontalisation system with little effort. The example code snippets from these sections can be combined into a fully working application with not many more lines than presented here. The pose estimation and landmark fitting run in a matter of milliseconds, while the texture remapping is the slowest component taking around 100 milliseconds (measured on an Intel Core i7-4700MQ). That is, to reduce dependencies of the framework and to make it run on any environment, it is not using OpenGL or any other acceleration technique. The main goals of the library are that it is simple, easy to use, easy to integrate into existing software and extensible. The full documentation, which includes additional functionality not presented here, as well as the complete and documented example application from this section, are available in the library repository.

4 CONCLUSIONS

We presented the *Surrey Face Model*, a multi-resolution 3D Morphable Face Model that is publicly available for non-commercial purposes. The

model is available in three different resolution levels and accompanied by isomap coordinates and landmark metadata. We introduced a lightweight header-only library written in modern C++ that accompanies the model and is actively developed on GitHub at `https://github.com/patrikhuber/eos`. The software features real-time shape model fitting and face frontalisation functionality and interoperability with OpenCV.

In contrast to existing work, the Surrey Face Model is available in multiple resolution levels and is built from racially diverse scans. Furthermore, a model-fitting software is available alongside the model to fit the model to novel images and videos. By designing the whole framework with simplicity as its foremost goal and using a public place for development and interaction, we hope to spur research with 3D Morphable Face Models in the community and encourage new parties to tackle their challenges with 3D face models. In addition to the full 3DMM being available via the University, we release a low-resolution shape model distributed directly within the public repository so that interested researchers can be ready-to-go in a matter of minutes.

Instructions to acquire the full model are available at `http://cvssp.org/facemodel`.

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